

SECTION A (40 Marks)

(Attempt *all* questions from this *Section*.)

Question 1

Select the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answer only).

[15]

- (i) The number of live births per thousand people per year is called:
- (a) Mortality
 - (b) Population density
 - (c) Population growth rate
 - (d) Natality
- (ii) The structure that lies at the junction of the choroid and the iris and is itself a part of the choroid is called:
- (a) Ciliary body
 - (b) Suspensory ligament
 - (c) Pupil
 - (d) Lens
- (iii) **Assertion (A):** Seminal vesicles pour an alkaline secretion into the semen as it passes through the ureter.
- Reason (R):** The female vagina is acidic and needs to be neutralized for sperm survival.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.
 - (c) Both (A) and (R) are true.
 - (d) Both (A) and (R) are false.
- (iv) The hormone that induces vigorous contractions of the uterus in a pregnant mother is:
- (a) Oestrogen
 - (b) Prolactin
 - (c) Oxytocin
 - (d) Progesterone

- (v) A scientist wanted to conduct a genetic experiment on mice. He cut the tails of all the mice and made them reproduce. What will be the phenotype of the offsprings?
- (a) All the offsprings will be tailless.
 - (b) There will be a mix of tailed and tailless offsprings.
 - (c) All the offsprings will have tails.
 - (d) Tail of the offsprings cannot be predicted.
- (vi) **Assertion (A):** A turgid plant cell is in a balanced state.
- Reason (R):** In a turgid cell, turgor pressure and wall pressure are equal and in the same direction.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.
 - (c) Both (A) and (R) are true.
 - (d) Both (A) and (R) are false.
- (vii) The optimum temperature for photosynthesis is:
- (a) 10 – 15°C
 - (b) 60 – 65°C
 - (c) 40 – 45°C
 - (d) 25 – 35°C
- (viii) Ajay and Vijay are siblings. Ajay can roll his tongue while Vijay cannot. What is the genotype of their parents?
- (a) Both the parents are homozygous dominant.
 - (b) Both the parents are homozygous recessive.
 - (c) One parent is homozygous dominant, and the other is homozygous recessive.
 - (d) Both the parents are heterozygous dominant.
- (ix) **Assertion (A):** Deficiency of Antidiuretic hormone causes an increase in urination.
- Reason (R):** Antidiuretic hormone acts on the kidney to reabsorb water from the kidney tubules.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.

- (c) Both (A) and (R) are true.
 - (d) Both (A) and (R) are false.
- (x) A phase of cell division where the spindle fibres disappear is:
- (a) Telophase
 - (b) Prophase
 - (c) Metaphase
 - (d) Anaphase
- (xi) **Assertion(A):** Arachnoid layer helps in speeding up the transmission of nerve impulses along the axons.
- Reason(R):** Axon endings at the synapse secrete acetylene.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.
 - (c) Both (A) and (R) are true.
 - (d) Both (A) and (R) are false.
- (xii) Many nuclear power plants are constructed near water sources. They use water for cooling their machinery and release it back into the water stream. Identify the kind of pollution caused.
- (a) Radiation pollution
 - (b) Water pollution
 - (c) Industrial pollution
 - (d) Thermal pollution

(xiii)



Owls are nocturnal animals. They are active at night and hunt small animals for food.

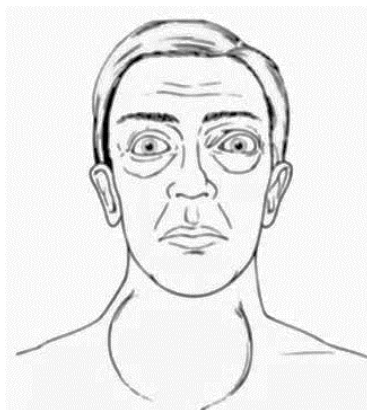
Owls have large eyes and can see well at night. This is due to:

- (a) Cones and Rhodopsin
 - (b) Rods and Iodopsin
 - (c) Rods and Rhodopsin
 - (d) Cones and Iodopsin
- (xiv) Sunil tried to match the tropic movements in plants with the stimulus they respond to. The tabulation is given below:

	Tropism	Stimulus
P	Thigmotropism	Touch
Q	Geotropism	Water
R	Chemotropism	Chemicals
S	Phototropism	Gravity

Identify the correct pair:

- (a) P and R
 - (b) Q and S
 - (c) R and S
 - (d) P and S
- (xv) Observe the following picture, identify the *defect* and the cause of the defect.



- (a) Simple goitre, under secretion of thyroxine.
- (b) Ex-ophthalmic goitre, over secretion of thyroxine.
- (c) Simple goitre, over secretion of thyroxine.
- (d) Ex-ophthalmic goitre, under secretion of thyroxine.

MARKING SCHEME

(i)	(d)	Natality
(ii)	(a)	Ciliary body
(iii)	(b)	A is false and R is true.
(iv)	(c)	Oxytocin
(v)	(c)	All the offsprings will have tails
(vi)	(a)	A is true and R is false
(vii)	(d)	25 – 35°C
(viii)	(d)	Both parents are heterozygous dominant.
(ix)	(c)	Both A and R are true.
(x)	(a)	Telophase
(xi)	(d)	Both A and R are false
(xii)	(d)	Thermal pollution
(xiii)	(c)	Rods and Rhodopsin
(xiv)	(a)	P and R
(xv)	(b)	Ex-ophthalmic goitre, over secretion of thyroxine

Comments of Examiners

- (i) Most of the candidates wrote the correct answer. However, 'mortality' / 'population growth rate' was written as an occasional incorrect answer.
- (ii) Majority of the candidates wrote the correct answer. However, a few candidates mentioned 'suspensory ligament' instead of 'Ciliary Body'.
- (iii) Several candidates could not answer it correctly. They interpreted the assertion statement to be true as they were not clear about the specific roles of each gland.
- (iv) Most of the candidates answered it correctly. However, a few candidates were confused and wrote 'oestrogen' or 'progesterone' instead of 'oxytocin'.
- (v) A large number of candidates were confused about the characters passing on to the next generation. They incorrectly chose option (d).
- (vi) A few candidates could answer this question correctly. Due to lack of clear understanding about turgor pressure, wall pressure and turgidity, most of the candidates incorrectly selected option (c) instead of option (a).
- (vii) All the candidates answered this question correctly.
- (viii) Most of the candidates wrote the correct answer. However, some candidates chose option (a) and option (c). They made random choices indicating their failure to associate phenotype with genotypes.
- (ix) Only a few candidates wrote the correct answer. Many selected the distractors as they were unable to relate the role of hormones with the function of the kidney tubules.
- (x) Most of the candidates answered this question correctly. However, a few candidates wrote 'Anaphase' in place of 'Telophase'.
- (xi) Only a few candidates chose the correct option. Most candidates wrongly selected option (a) or option (b) indicating inadequate understanding of the concept.
- (xii) Majority of the candidates answered this question correctly. However, some candidates opted for 'radiation pollution' / 'water pollution'.
- (xiii) Majority of the candidates wrote the correct answer. However, few candidates

Suggestions for teachers

- Teach natality, mortality, and population growth rate through step-by-step breakdowns of population change scenarios. Follow up with targeted MCQs that test subtle differences among the terms.
- Show location and function of both ciliary body and suspensory ligaments by using 3D models, charts, diagrams and videos to bring clarity.
- Put stress on the role of the ciliary body in accommodation of eye.
- Clearly explain the function of each male accessory gland (seminal vesicles, prostate gland, and Cowper's glands), emphasising their contribution in semen formation.
- Use comparative tables and diagrams to reinforce differences among the glands.
- Discuss clearly the roles of different hormones in reproduction.
- Highlight the function of oxytocin in parturition and milk ejection.
- Reinforce principles of inheritance using examples where dominant/recessive traits are visible in phenotypes.
- Teach how to evaluate assertions and reasons critically and ensure sufficient practice of the same.
- Provide a clear understanding of the concepts of turgor and wall pressure along with examples from daily life experiences.
- Demonstrate turgor pressure by comparing a wilted and a healthy plant in class / in laboratory.
- Elucidate explicitly the role of temperature in photosynthesis citing some real-life examples like reduction in the rate of photosynthesis during cold or heatwaves.
- Provide more examples other than that discussed in the textbook on genetic crosses, for better understanding and retention of the concept of inheritance.
- Encourage students to use of Punnett squares to deduce parent genotypes from offspring outcomes.
- Familiarise students with the vocabularies like dominant, recessive, heterozygous, homozygous, etc. with deep conceptual understanding using suitable examples.

were confused between the functions of the sensory cells in dim light and bright light and opted for the wrong answer.

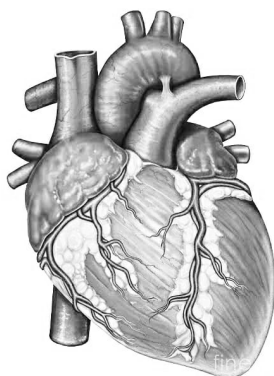
- (xiv) All the candidates answered this question correctly.
- (xv) A large number of candidates wrote the correct answer. However, some candidates failed to relate the given defect with the specific hormone and lost marks.

Suggestions for teachers

- Advise students to read the statements carefully to identify incorrect scientific statements while solving Assertion - Reason type question.
- Conduct group discussion on various case studies for effective understanding of concepts related to hypo or hypersecretion of hormones.
- Discuss the events of mitosis highlighting the unique events and their importance, using video, animations, diagrams or stage-wise cell division charts.
- Emphasise on the usage of correct spellings for biological terms.
- Provide clear understanding of the types of pollution with relatable real-world examples and case studies for each type.
- Reinforce the function and pigments of rods and cones. Relate rhodopsin to dim light vision through activities or examples from daily life.
- Clearly explain the types of tropic movements (phototropism, geotropism, etc.) and the specific stimuli involved.
- Conduct laboratory activities using a clinostat to demonstrate plant responses to light or gravity and observe root movements towards water to put stress on these concepts.
- Summarise the disorders due to hormone imbalances (hypo or hypersecretion) using comparison tables comprising the cause, hormone involved, and symptoms and share it with students while teaching endocrine system.
- Instruct students to observe the image carefully before answering image-based questions.

Question 2

- (i) Name the following: [5]
- (a) The fully developed part of the ovary containing a mature ovum.
 - (b) A waxy layer on the upper epidermis of leaves meant to reduce Transpiration.
 - (c) The ear ossicle which opens into the oval window of Cochlea.
 - (d) The physiological process which is the starting point of all food chains.
 - (e) An insect which is a classic example of natural selection due to industrial melanism.
- (ii) Given below is the diagram of a human heart. Read the information below the diagram and fill in the blanks: [5]



The human heart pumps blood throughout the body. It is the size of a large fist. The heart is located between the lungs in the thoracic cavity. It has four chambers. The heart functions all through the lifespan of a person and is responsible for the survival of the person.

- The heart is enclosed by a membrane called (a) _____. The ventricles give rise to two large blood vessels called (b) _____ and (c) _____. The flaps of the cuspid valves are kept in position by (d) _____. (e) _____ arteries supply oxygenated blood to the walls of the heart.
- (iii) Arrange the terms in each group in the correct order. Write them in a logical sequence beginning with the term that is **underlined**. [5]
- (a) Palisade cells, Cuticle, Upper epidermis, Spongy cells.
 - (b) Hepatic portal vein, Stomach, Hepatic vein, Liver.

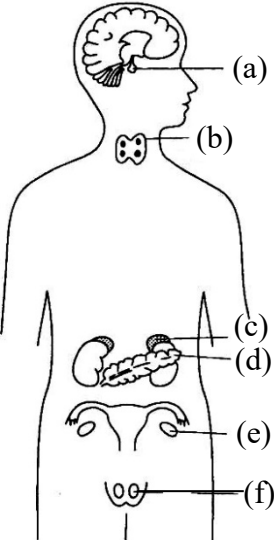
- (c) Cyton, Dendrites, Synaptic cleft, Axon endings.
- (d) Insulin release, High blood sugar, Lowers blood sugar, Stimulates glucose uptake from blood.
- (e) Sperm duct, Seminiferous tubule, Epididymis, Efferent ducts.
- (iv) Read the explanations given below and name the structures: [5]

Example: *The tubules which produce sperms.*

Answer: *Seminiferous tubules*

- (a) The canal through which the testes descend down into the scrotum just before the birth of a male baby.
- (b) The part of the brain that regulates the peristaltic movement of the alimentary canal.
- (c) A structure that collects the soundwaves and conducts them through the external auditory canal.
- (d) The point of attachment of two chromatids of a chromosome.
- (e) Regulation of the opening and closing of stomata.
- (v) Given below is an outline of the human body showing the important endocrine glands. Match the glands marked (a) to (e) with their correct functions: [5]

Example: (f) – 6. Secretes Testosterone.

Endocrine glands	Functions
	1. Prepares the body to face stress.
	2. Secretes tropic hormones
	3. Secretes Oestrogens
	4. Maintains blood sugar level
	5. Regulates basal metabolism
	6. Secretes Testosterone.

MARKING SCHEME

(i)	(a) Graafian follicle/ mature ovarian follicle / tertiary follicle (b) Cuticle/ Cuticular layer (c) Stapes / Stirrup (d) Photosynthesis (e) Peppered moth / <i>Biston betularia</i>
(ii)	(a) Pericardium (b) Pulmonary artery or aorta (c) Aorta or Pulmonary artery (d) Chordae tendineae (e) Coronary
(iii)	(a) Cuticle, Upper epidermis, Palisade cells, Spongy cells. (b) Stomach, hepatic portal vein, liver, hepatic vein. (c) Dendrites, cyton, axon ending, synaptic cleft. (d) High blood sugar, insulin release, stimulates glucose uptake from blood, lowers blood sugar. (e) Seminiferous tubule, efferent ducts, epididymis, sperm duct.
(iv)	(a) Inguinal canal (b) Medulla oblongata /Brainstem (c) Pinna/Auricle /Auricula (d) Centromere (e) Guard cells
(v)	(a) 2. Secretes tropic hormones (b) 5. Regulates basal metabolism (c) 1. Prepares the body to face stress. (d) 4. Maintains blood sugar level (e) 3. Secretes Oestrogens.

Comments of Examiners

- (i) (a) Some candidates wrote 'ovum/ follicle' instead of "graafian follicle" for the fully developed part of the ovary.
- (b) Some candidates wrongly mentioned 'cuticular transpiration' in place of 'cuticle', leading to loss of marks.
- (c) Several candidates were confused with the sequence of the ear ossicles in the middle ear. Instead of 'stapes', other ear ossicles like 'malleus' or 'incus' were referred by them.
- (d) Some candidates incorrectly wrote 'respiration in plants' instead of 'photosynthesis' indicating lack of clarity about the two vital processes.
- (e) Some candidates mentioned 'moth/ paper moth/ industrial moth/ black moth/ moth-Biston' instead of "Peppered Moth/ *Biston betularia*." Spelling errors such as 'paper moth' was also noted in few answer scripts.
- (ii) (a) Some candidates wrote 'pleura' or 'pericardial fluid' instead of 'pericardium'.
- (b) Many candidates were confused between 'pulmonary artery' and 'vein'. Use of the short forms like 'P.A', 'Pul. Art' by the candidates led to the loss of marks in many instances.
- (c) Many candidates mentioned 'vena cava' instead of 'aorta'. Some candidates misspelled 'aorta' as 'arota'.
- (d) Some candidates skipped this part of the question or incorrectly referred 'valves/ papillary muscles' instead of 'chordae tendineae'. Spelling errors were noticed in several answer scripts.
- (e) Several candidates wrote the correct answer. However, some candidates wrote 'pulmonary arteries' instead of 'coronary arteries'.
- (iii) (a) Several candidates failed to envision the correct anatomical layering of a leaf. Some of the candidates were confused regarding the positions of palisade and spongy cells.
- (b) A large number of candidates failed to score as they changed the sequence of 'hepatic vein' with 'hepatic portal vein'. Some of the candidates wrongly placed 'hepatic portal vein' before 'stomach'.

Suggestions for teachers

- Lay emphasis on the stages of follicular development focusing on the terminology used for different stages.
- Use clear diagrams or 3D animations to show the transition from primordial follicles to Graafian Follicles.
- Clarify the difference between the ovum and the Graafian Follicle and expose the students to wide variety of questions based on the concept.
- Explain elaborately the changes observed during different phases of the menstrual cycle with visual stimuli like diagram/ flowchart or 3D animations.
- Highlight the role of the waxy cuticle in reducing water loss and emphasise the key differences between the cuticle and epidermis.
- Offer microscopic observation of the structure of the leaf along with a lucid diagram for the concept clarity of the students.
- Teach ear ossicles as per their order, with the help of flow charts/diagrams or 3D animations to build better understanding.
- Bring out clear differences between the vital plant physiological processes like photosynthesis and respiration by a comparative analysis.
- Discuss the role of environmental changes (such as pollution) in selecting advantageous traits and reinforce the examples of natural selection.
- Present different case studies like industrial melanism, specifically using the example of the peppered moth for better retention of the concept.
- Ensure that the students write the correct scientific names following the principle of binomial nomenclature.
- Explicitly elucidate the anatomical layers of the heart using 3D models or interactive visuals.
- Highlight the terms like pericardium (membrane of heart), pleura (lungs), and pericardial fluid (for lubrication) to bring out the contrast among these similar sounding words.
- Spell out the difference between arteries and veins using flowcharts showing blood circulation through the heart.

- (c) Some of the candidates were confused with the order of neural transmission, specially with the position of axon endings and synaptic cleft.
- (d) Many candidates failed to comprehend the logical endocrine sequence and presented random orders.
- (e) Several candidates were confused with the sequence of structures present in the male reproductive system.
- (iv) (a) Some candidates were not clear about the terms like 'inguinal canal', 'sperm duct' and 'scrotum'. Spelling errors of 'inguinal canal' were committed by many.
- (b) Only few candidates were able to write the correct answer. Many candidates were confused and wrongly mentioned 'cerebellum' instead of 'medulla oblongata'.
- (c) Majority of the candidates answered this part correctly. However, a few candidates wrote 'external auditory canal' instead of 'pinna'.
- (d) Many candidates were not clear about the difference among the similar sounding terms like centromere/chromatid/chromosome/centrosome. They failed to mention centromere leading to the loss of marks.
- (e) Most of the candidates correctly wrote the answer. However, spelling errors were observed in some answer scripts. Few candidates mentioned 'bean shaped cells' only instead of 'guard cell'.
- (v) Many candidates did not follow the correct numbering pattern which lead to confusion while interpreting the answers. Due to the inability to correlate the endocrine glands with the hormones produced a large number of candidates failed to score in this part.

Suggestions for teachers

- Use visual aids to highlight the structure and function of the chordae tendineae. Include simple animations or labelled diagrams to show how they prevent valve prolapse.
- Emphasise that coronary arteries supply oxygenated blood to the heart muscles. Use clinical examples (like angina) to relate the function to real-life context and aid retention.
- Ensure students understand the sequence of blood flow logically. Provide adequate practice to the students encouraging them to repeatedly write the important terms in correct sequence.
- Teach male reproductive system using charts or 3D models or video animation.
- Explain the basic functions of cerebrum, cerebellum and medulla oblongata correlation daily life activities for better understanding.
- Use ear model kits or conduct a class skit where sound travels from one structure to another; stress that the pinna collects sound, not conducts.
- Provide printed chromosome diagrams with movable parts; ask the students to physically join chromatids at the centromere to understand structure and terminology.
- Ensure students learn and correctly differentiates all the similar sounding words like centromere/ centrosome, Chromatid/ chromosome.
- Discuss the mechanism of opening and closing of stomata with diagram, chart and appropriate animation.
- Emphasise the location and function of guard cells by offering microscopic observation to the students.
- Advise them to connect hormones with daily life examples (e.g., adrenaline when stressed, insulin after eating sweets).

SECTION B (40 Marks)

(Attempt *any four* questions from this *Section*.)

Question 3

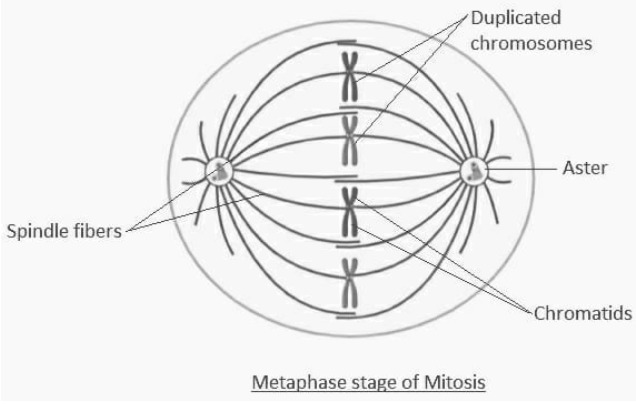
- (i) Write the overall chemical equation for photosynthesis. [1]
- (ii) Ramesh is a farmer who has a vineyard and cultivates grapes. To improve the shape of the grapes and make them elongated he sprays a particular *phytohormone* at the time of fruiting. [2]



- (a) Name the *phytohormone* that Ramesh sprays on the plant at the time of fruiting.
- (b) These grapes are said to be seedless. What is the technical term for the formation of such fruits?
- (iii) Mention the *two* pairs of nitrogenous bases that are present in a DNA strand and pair with each other by hydrogen bonds. [2]
- (iv) State *any two* disadvantages of living close to an airport. [2]
- (v) Draw a neat, labelled diagram of an animal cell showing Metaphase with four chromosomes. [3]

MARKING SCHEME

(i)	$6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow[\text{Chlorophyll}]{\text{Sunlight / light / solar energy / Photons}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$
(ii)	(a) Gibberellins/Gibberellic acid/ GA / GA3 (b) Parthenocarp
(iii)	Adenine – Thymine Cytosine – Guanine

(iv)	- Lowers efficiency of work - Disturbs sleep , leads to fatigue - Damages ear drum, irritates lungs - Can lead to deafness, hearing problems - Lowers efficiency of work, causes depression, causes anxiety - High blood pressure, increased pulse rate - Leads to stress, heart attack, irritability - Speech interference, stress - High BP, heart ailment, risk of stroke - Noise pollution, Air pollution - Traffic congestion <i>(Any two)</i>
(v)	 <u>Metaphase stage of Mitosis</u>

Comments of Examiners

- (i) Most candidates answered this question correctly. However, some missed one of the products, such as water, or wrote an unbalanced equation. A few candidates also failed to mention one or both of the conditions required for photosynthesis.
- (ii) (a) Majority of the candidates wrote the answer correctly. However, a few candidates mentioned 'ethylene', as they misinterpreted the phytohormone in terms of fruit ripening.
(b) Many candidates could not recall the term 'parthenocarpy'. They confused it with 'pollination' / 'fertilisation'.
- (iii) Most candidates answered it correctly. However, some candidates, due to a lack of clarity about base pairing, were confused and presented incorrect pairing of the nitrogenous bases. A few candidates misspelt 'Thymine' as 'Thiamine' and 'Adenine' as 'Adenosine'.
- (iv) Many candidates wrote some random answers like 'it is noisy' or 'pollution' without specifying 'noise pollution' or 'sleep disturbance'. Some candidates supplied vague answers, such as 'tall buildings cannot be built', 'windows shake', and 'planes crash', etc.
- (v) Some common errors observed were as follows:
 - Supplying incomplete diagrams.
 - Drawing a plant cell instead of an animal cell.
 - Failure to show the duplicated chromosomes.
 - Inability to present chromosomes at the equatorial plane.
 - Absence of centrosome/centrioles in the diagram.
 - Missing the labelling.
 - Presenting Anaphase stage instead of Metaphase.

Suggestions for teachers

- Reinforce the equation of photosynthesis through repeated drills. Ensure that the students mention both the conditions of sunlight and chlorophyll above and below the arrow.
- Teach students to connect the equation to the type of reactions discussed in chemistry chapters (oxidation, reduction, substitution etc.)
- Reiterate that water needs to be mentioned both in the reactant as well as product side.
- Emphasise on balancing the equation.
- Summarise the phytohormones, their functions along with examples in a simple tabular form.
- Familiarise students with the key terminologies through regular vocabulary reviews and flashcards.
- Use fruit samples or images to relate technical terms to real-world examples like seedless papayas, oranges, etc.
- Show color-coded DNA strand models in class or use animation videos to elucidate explicitly the structure of DNA and its base-pairing rules.
- Discuss with students the real-life conditions of living near airports with the help of different case studies.
- Demonstrate step-by-step drawing of the stages of mitosis, emphasising the main changes observed during each stage.
- Advise students to show minimum essential labels in the answer script, as expected by examiners.
- Ensure students show the same number of chromosomes in the drawing as given in the question.
- Ask students to formulate questions based on these concepts, to make them think deeper w.r.t these concepts.

Question 4

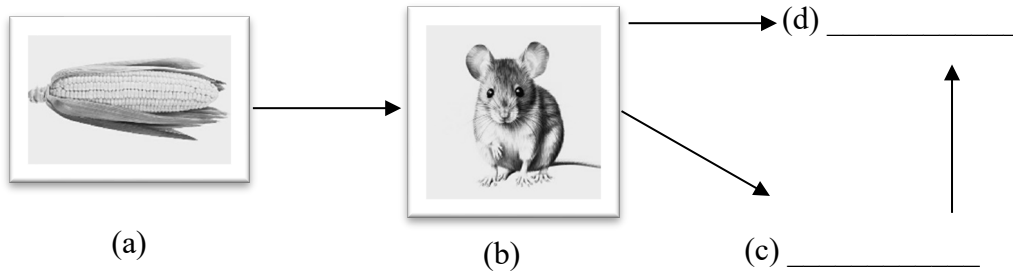
(i) What is the structure of DNA as proposed by Watson and Crick? [1]

(ii) Name the white blood cells that: [2]

(a) show phagocytosis

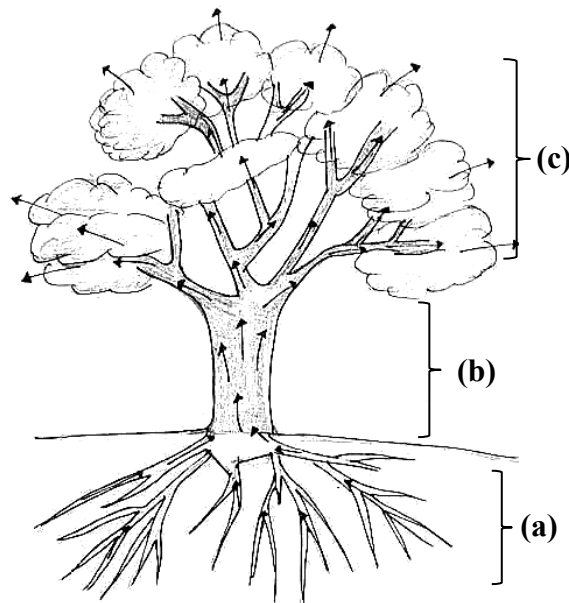
(b) produce antibodies

(iii) Complete the food chain by mentioning the names of the organisms in (c) and (d): [2]



(iv) State *any two* harmful effects of acid rain. [2]

(v) Given below is a picture of a tree. Mention the scientific phenomena occurring in the parts labelled (a), (b) and (c). [3]



MARKING SCHEME

(i)	Double helix / double helical structure
(ii)	(a) Neutrophils/ Monocytes / Macrophages / mononuclear leukocytes / B Lymphocytes / Eosinophils (b) Lymphocytes
(iii)	(c) Snake / Cat / mongoose / eagles / hawks / owls / foxes / large lizards (d) Hawk / Falcon / Kite / Eagle / Vulture/bears / wolves eat young ones of eagles and hawks
(iv)	- Damage to plants, building materials, paints, monuments, statues, marble - Harms aquatic life, forests, leaves - Water bodies, soil become acidic, Water Pollution, decrease in agricultural production - Corrodes steel bridges, stone buildings - Harms human skin, eyes, lungs - Fertility of soil reduced. - Alters PH of soil - Leaching of soil nutrients <i>(Any two)</i>
(v)	(a) Absorption of water, endosmosis (b) Ascent of sap / conduction, transport of water (c) Transpiration or explanation

Comments of Examiners

- (i) Instead of 'helical structure', many candidates wrote 'spiral structure'. Some of the candidates have explained the structure of DNA with a diagram without mentioning the 'double helical structure'. Additionally, a few candidates vaguely wrote 'ladder', 'twisted ladder', etc.
- (ii) Many candidates were confused about the types of WBCs and their corresponding functions. They wrongly wrote 'lymphocytes' instead of 'neutrophils'. Few candidates wrote 'WBCs' or 'leukocytes' in general instead of specifying the type of WBC.
- (iii) Majority of the candidates wrote the correct answer. However, a few wrote 'insects' in both the blanks.
- (iv) Almost all candidates answered this question correctly. Though vague responses like 'pollution' or 'harms plants' instead of specific effects like 'corrosion of monuments' or 'leaching of nutrients' were found in some answer scripts.
- (v) Most of the candidates answered this part of the question correctly. However, some candidates failed to identify the three phenomena.

Suggestions for teachers

- Highlight the keywords like 'double helix', 'complementary base pairing', and 'anti-parallel strands' through short recaps, charts, and classroom models.
- Show the 'Ladder like appearance of DNA strand' using a 3D animation and explain clearly for the better understanding of the students.
- Provide simple comparison tables of the types of WBCs and their functions and conduct frequent oral quizzes for better retention.
- Offer engaging classroom activities like solving food chain puzzles or finding the missing link in a food chain on flashcards.
- Put stress on the rule: Producer → herbivore → carnivore → top carnivore and the idea of 'eating and being eaten'.
- Teach the effects of environmental pollutions using visual examples e.g., photos of Taj Mahal damage, dying forests etc.
- Clearly elucidate each environmental issue using bullet points and ready-reference phrases while teaching in the class.
- Explain the physiological phenomenon occurring in each part of the plant body with classroom or laboratory demonstration (if possible) or with diagrams to avoid confusion.
- Provide regular practice of diagrams with review.

Question 5

- (i) What is Micturition? Explain. [1]
- (ii) What are biomedical wastes? Give an example. [2]
- (iii) Differentiate between the functions of Urethra and Uterus. [2]
- (iv) Rachna is an expert gymnast on the balance beam. She has been vigorously practising for the Olympic Games. Lately, she has been experiencing dizziness and losing her balance during the practise sessions. [2]

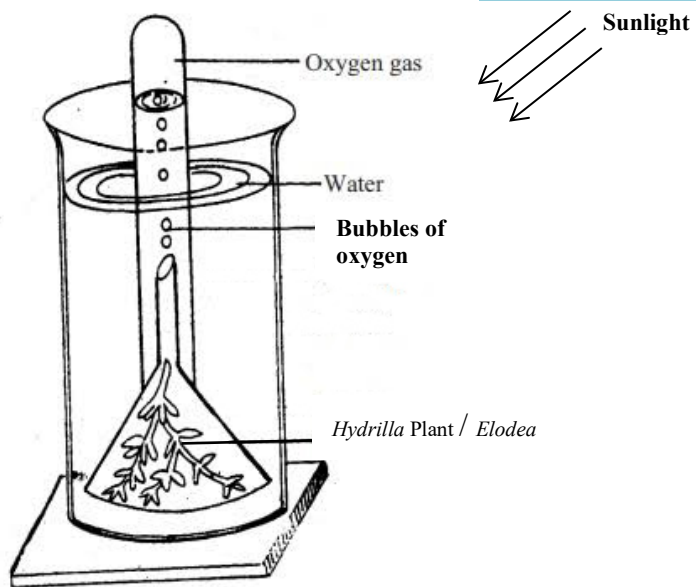


- (a) Which part of the membranous labyrinth is responsible for maintaining the body balance on the beam?
- (b) Which type of balance is maintained by the structure mentioned in (a)?
- (v) Draw a neat, labelled diagram to show that oxygen is released during photosynthesis. [3]

MARKING SCHEME

(i)	Elimination of urine from the body /passing urine from the body
(ii)	Any waste generated in hospital, clinic or lab, research labs, diagnostic centres is called biomedical waste. E.g-needles, syringes, dirty dressing, discarded medicinal tablets etc., sharp objects, used bandages, body fluids, human and animal tissues, body parts, placenta, Umbilical cords, discarded gloves, expired drugs, Hg from broken thermometers, used glass slides, used swabs, etc.
(iii)	Urethra – Eliminates urine from the urinary bladder to outside the body. Uterus – Houses the embryo, implantation of embryo, protects embryo – provides nourishment to embryo – provides suitable environment for foetal development – rhythmic contraction to transport sperms to oviduct – contracts leading to childbirth – sheds lining during menstruation
(iv)	(a) Semicircular canals/ vestibules / Utriculus / Sacculus (b) Dynamic balance/ equilibrium / static balance / equilibrium

(v)



Comments of Examiners

- (i) Most of the candidates wrote the correct answer but a few candidates merely wrote 'passing urine' without explaining the role of the urinary bladder and sphincter muscles. Some candidates wrote 'removal of wastes from the body' and lost marks.
- (ii) Most of the candidates wrote the answer correctly. However, some candidates wrote vague answers like 'waste from medicals' without including specific terms such as hospitals/laboratory/dispensary and provided vague examples like 'plastic' instead of 'used syringes' or 'body tissues'. A few candidates were confused between biomedical wastes and biodegradable wastes and failed to score.
- (iii) Many candidates interchanged the functions or wrote anatomical locations instead of functions. Some candidates were confused between 'uterus' and 'ureter' leading to the loss of marks.
- (iv) (a) Majority of the candidates failed to understand that the answers for parts (a) and (b) are linked. They wrote incorrect answers like 'semicircular canals and static equilibrium', 'vestibule and dynamic equilibrium'.
- (b) Most of the candidates were unable to relate the type of balance to the structure mentioned in (a).
- (v) Some common errors made by the candidates were as follows:
 - Poor drawing of diagrams.
 - Missing labels.
 - Merging the funnel and test-tube in the diagram.
 - Not showing 'oxygen release' from the test tube.
 - Wrongly showing the level of water below the stem of the funnel in the beaker.
 - Not showing the Sunlight/ light rays.

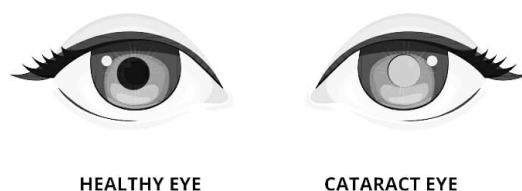
A lot of unrelated diagrams were drawn by many candidates. For example, a leaf showing the intake of CO₂ and releasing O₂, stomatal apparatus, chloroplast etc.

Suggestions for teachers

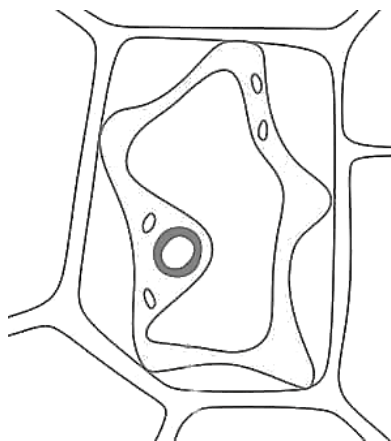
- Ensure students learn definitions related to human physiology with keywords of the specific organ systems involved.
- Provide clear and consolidated understanding of the 'role of urinary bladder and urethra during the expulsion of urine from the body' with the help of labelled diagrams or short video clips.
- Use real-world hospital examples of biomedical wastes and perform 'sort-it-out activities' with mock items to overcome confusion among students.
- Bring out the structure-function comparisons using Venn diagrams or two-column charts. Reinforce through pair-share activities and formative quizzes.
- Clearly elucidate inner ear anatomy using labeled 3D models or virtual anatomy apps.
- Emphasise on the role of each part of ear through movement-based analogies (e.g., dancing, gymnastics).
- Demonstrate the classic *Hydrilla* experiment in class/laboratory.
- Draw/display sample diagrams on the board/smart board and advise students to practice diagrams regularly.

Question 6

- (i) Where are Hydathodes located? [1]
- (ii) What are Vestigial organs? Give *one* example of a vestigial structure in the human mouth. [2]
- (iii) Ahmed had a severe throat infection. After a few days he developed an earache too. What could be the reason for the pain in his ears? Explain. [2]
- (iv) Mr. Sharma is a 76-year-old man. His vision was gradually becoming blurred. He consulted an eye doctor and it was diagnosed as cataract. [2]



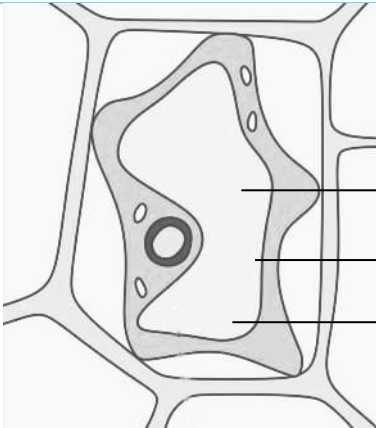
- (a) Which part of the eye does cataract affect?
- (b) Why does the vision become blurred?
- (v) Given below is the diagram of a *plasmolysed* cortical cell of a root. [3]



Copy the diagram and label the following parts in the diagram:

- (a) Cytoplasm
- (b) Hypertonic solution
- (c) Plasma membrane

MARKING SCHEME

(i)	At the ends of veins, margins of leaves / tips of leaves, coleoptiles, tendrils
(ii)	Organs / structures which have lost their functions , / and exist in a reduced form Wisdom teeth/ tooth
(iii)	– Ear and throat are connected by the eustachian tube. – Germs can travel from throat to ear and cause the ear ache. – Irritation of nerves connecting ear and throat.
(iv)	(a) Lens (b) Lens becomes opaque
(v)	 Cytoplasm / (a) Hypertonic solution / (b) Plasma membrane / (c)

Comments of Examiners

- (i) Most candidates answered it correctly. However, some candidates simply wrote 'leaves' instead of 'margins or tips of leaves.' Few candidates were confused between 'hydathodes' and 'stomata'.
- (ii) Due to lack of clarity, several candidates confused vestigial organs with non-essential organs. Many candidates wrote 'teeth/ molar teeth' instead of 'wisdom teeth/ 3rd molar.' Some candidates overlooked the phrase 'in the mouth' and gave examples of other unrelated vestigial organs like 'pinna', 'appendix' etc. Few candidates used inappropriate expression to define vestigial organs as 'useless body parts'.
- (iii) Only a few candidates could answer this part correctly. Majority of the candidates failed to mention 'Eustachian tube' as the tube connecting the throat and middle ear, often attributing to the pain to cold or general infection. Many candidates failed to mention that the pathogens travelled from the throat to the ear.
- (iv) (a) Many candidates answered this sub-part of the question correctly. However, some candidates incorrectly wrote 'cornea', 'retina' or 'iris' instead of 'Eye lens' and failed to score.
(b) Some candidates gave vague answers like 'the eye is weak' rather than mentioning 'loss of lens transparency/ lens becomes opaque'. Additionally, few candidates mentioned that the 'Cornea / Pupil / Iris becomes opaque'.
- (v) (a) Majority of the candidates failed to answer it correctly. Diagram was copied correctly by most of them, but the labels were misplaced or inaccurate. Some candidates labelled the nucleus as 'cytoplasm.' Cytoplasm was wrongly shown between the cell wall and cell membrane by a few candidates.
(b) Most candidates marked the 'cell sap' of the vacuole as the 'hypertonic solution'.
(c) Many candidates incorrectly marked the cell wall as the plasma membrane.

Suggestions for teachers

- Display comparative visuals of stomata vs. hydathodes using diagrams/ digital diagrams/charts or 3D animations.
- Provide regular practice of location-based identification with close-up images.
- Clarify definitions using evolutionary context. Present everyday examples (e.g., goosebumps, appendix, wisdom teeth) for better retention.
- Organise classroom quizzes on "myth or fact" to correct misconceptions regarding the human body among the students.
- Train the students to read the complete question carefully and answer accordingly.
- Show the link between throat and ear with a labelled side-view diagram or 3D model of head anatomy.
- Acquaint students with the interconnections of ENT systems with the help of 'Role-play' or 'storytelling' techniques.
Examples – nose watering while crying inconsolably, bitter taste of medicines in the mouth when one puts eyedrops in the eyes.
- Explain eye anatomy using sectional diagrams and 3D images or animations.
- Put stress on different age-related eye disorders through short case-based discussions.
- Encourage students to connect image formation in eye to the ideas discussed in physics about image formation in lens.
- Provide model-based explanations of transparent vs. opaque visuals. Encourage use of keywords like 'clouding' and 'opacity.'
- Bring out the contrast between the abnormalities of cornea and lens.
- Conduct step-by-step drawing sessions with peer assessment. Use actual microscope images of plasmolysis for reference and better understanding.
- Clearly elucidate the concept of tonicity, and highlight the changes observed in the cell when kept in a hypertonic/hypotonic solution.

Question 7

- (i) Define the term Allele. [1]
- (ii) In the last century, the human population has increased tremendously. Death rate has reduced and life expectancy has increased. This rapid growth has placed significant pressure on resources. Global cooperation is needed for environmental preservation. [2]



- (a) Mention *any one* reason for reduction in the death rate.
- (b) Name *any one* resource that is under pressure due to population explosion.
- (iii) Mention *any two* functions of amniotic fluid. [2]
- (iv) Expand the following abbreviations: [2]
- (a) CFC
- (b) CNG
- (v) Vineet's day begins with a cup of piping hot coffee. Caffeine is a stimulant present in coffee which increases the circulation of Adrenaline in the body. Vineet feels refreshed, alert and focussed. [3]



- (a) Name the organ that is stimulated by caffeine which makes Vineet alert and focussed.
- (b) Give the collective term for the protective membranes of the organ mentioned in (a).
- (c) Which part of the autonomic nervous system increases the secretion of Adrenaline?

MARKING SCHEME

(i)	Alternative forms of a gene/ different forms of a gene, /1 or more possible forms of a gene
(ii)	(a) – More medical facilities/ access to hospitals – Spread of education/ awareness of hygiene, quarantine measures – Control of epidemics/ diseases – Balanced diet – Advancement in Science and technology/ development of Antibiotics – Improved standard of living/ sanitation – Mass immunisation, vaccinations – Better food storage – Food shortages minimized (Any one) (b) Food, Land, Water, Forests, Fossils fuels, minerals, wildlife, petroleum products
(iii)	– Maintains even pressure, temperature around the foetus/ baby – Protects foetus / baby from injuries, mechanical shocks – Keeps foetus moist – Allows foetal movement – Prevents sticking of foetus to amnion (Any two)
(iv)	(a) CFC – Chlorofluoro Carbon CNG – Compressed Natural Gas
(v)	(a) Brain, cerebrum, cerebral hemispheres (b) Meninges (c) Sympathetic nervous system

Comments of Examiners

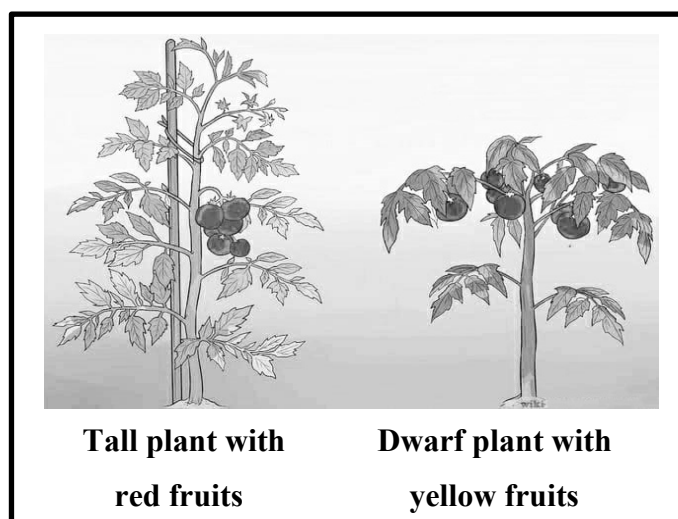
- (i) Many candidates wrote vague definitions such as ‘form of a gene/ forms of traits or characters’ without specifying that alleles are alternative forms of a gene at the same locus. Few candidates explained it as Dominant or Recessive traits.
- (ii) (a) Some candidates mentioned improved “technology” or “economy” instead of mentioning the specific reasons like ‘advanced medical care’ or ‘vaccination’. ‘Increase in birth rate’ was also found in few answer scripts.
(b) Most of the candidates gave vague answers like ‘everything’ or ‘oxygen’.
- (iii) Many Candidates were confused between ‘amniotic fluid’ and ‘placenta’. Some candidates supplied wrong answer like ‘nutrient supply.’
- (iv) (a) Majority of the candidates answered correctly. However, several candidates expanded CFC incorrectly as ‘compressed fluorocarbons’.
(b) Many candidates were confused between CNG and LPG. Some candidates used several self-created terms like “Carbon Neutral Gas / Compressed Nature Gas / Central Natural Gas” and lost marks.
- (v) (a) Only a few candidates answered it correctly. Some candidates wrote ‘Adrenal Gland / Adrenal medulla’ instead of writing the correct organ- ‘brain’.
(b) Majority of the candidates mentioned the three layers covering the brain instead of coming up with the correct collective term ‘meninges’.
(c) Several candidates failed to recall the term ‘sympathetic nervous system’ or confused it with ‘parasympathetic’. Many candidates came up with wrong answers like CNS / PNS / ANS.

Suggestions for teachers

- Reinforce definitions using examples of Punnett square. Use commonly observed traits (like eye colour) to demonstrate allele variations.
- Encourage students to conduct a survey or identify examples like family history, traits, etc. that could be converted to cases to strengthen the ideas of genotype-phenotype links through Punnett square activities.
- Focus on cause-effect charts showing population vs. healthcare advancements. Use timelines to show the impact of vaccines.
- List and discuss natural resources under pressure using infographics. Discuss local case studies (e.g., water shortage in cities) for better understanding.
- Include “match the resource with problem” activities while teaching population.
- Explain fetal environment with the help of a 3D fetal model or videos.
- Provide fill-in-the-blank type practice questions on functions of placenta vs. amniotic fluid.
- Spell out the differences between amniotic fluid, placenta and umbilical cord clearly. Ensure practice of similar questions in class.
- Teach expansions of the common abbreviations using flashcards or acronym games.
- Use comparison tables between CNG, LPG, and petrol to familiarise students with all these terms. Introduce transport posters or ads for CNG vehicles in class to make the lesson more engaging.
- Highlight brain’s role in reflexes and alertness.
- Put stress the function and names of protective coverings with mnemonics. Collective term- Meninges.
Mnemonic: "D-P-A: Don't Push Anyone"
This stands for:
 - D – Dura mater (outermost, tough and durable)
 - P – Pia mater (innermost, delicate and closely associated with the brain)
 - A – Arachnoid mater (middle, web-like layer)
- Teach using analogies (e.g., “sympathetic speeds you up; parasympathetic powers you down”) to overcome confusion among students.

Question 8

- (i) Explain the term Excretion. [1]
- (ii) State the difference between *Nephron* and *Neuron*. [2]
- (iii) Mention *any two* objectives of Swachh Bharat Abhiyan. [2]
- (iv) Give *any two* characteristic features of a Cro-Magnon man. [2]
- (v) A farmer cultivates a unique species of tomato plants. The two traits selected for breeding were: Colour of the fruits and height of the plants. The farmer initially crosses a homozygous tall plant bearing red fruits (**TTRR**) with a dwarf plant bearing yellow fruits (**ttrr**) and the genotype of all the offsprings of the **F₁** generation was **TtRr**. Next, he crosses two **F₁** generation plants (**TtRr**). [3]



- (a) Explain the phenotype of the offsprings of the **F₁** generation obtained from the initial cross.
- (b) Mention the phenotypic ratio of the **F₂** generation offsprings.
- (c) State Mendel's Law of Independent Assortment.

MARKING SCHEME

(i)	It is the elimination of nitrogenous wastes / urea from the body
(ii)	Nephron – Structural and functional unit of kidneys Neuron – Structural and functional unit of nervous system
(iii)	– Clean streets, roads, cities, towns – Eliminate open defecation, promoting sanitation, hygiene – Construction of latrines/ toilets, monitoring toilet use

	– Efficient management of solid and liquid wastes – Make India clean – Proper waste management – Cleaner environment <i>(Any two)</i>
(iv)	– perfect bipedalism – small canines – well developed chin – reduced body hair – broad forehead, large teeth, large skull – cranial capacity 1600 cc, larger brains – prominent browridges – Tall – 5.5' – 5.7' – well built body <i>(Any two)</i>
(v)	(a) All are tall plants bearing red fruits (b) 9 : 3 : 3 : 1 (c) When there are two pairs of characters, the distribution of alleles of one character is independent of the distribution of the alleles of the other character into the gametes. Genes for different traits are distributed into gametes independently. Alleles of 2 different genes are sorted into gametes independently.

Comments of Examiners

- (i) Most candidates were able to answer it correctly. However, some candidates were confused with the terms 'excretion', 'egestion' and 'defecation'. Additionally, a few candidates failed to specify that excretion is the process of removing nitrogenous waste material from the body.
- (ii) Most candidates answered this sub-part of the question correctly.
- (iii) Majority of the candidates wrote vague objectives like 'to keep clean' without relating them to actual goals like sanitation or waste disposal.
- (iv) Many candidates wrote generic traits like 'used tools' or 'were intelligent' without mentioning the specific features like 'broad nose' or 'prominent chin'.
- (v) (a) Most of the candidates answered it correctly. However, some candidates wrote 'tall red' without justifying the hybrid nature or simply 'Tall'.
 (b) Majority of the candidates answered it correctly. However, some common errors observed were:
 - writing the incorrect phenotypic ratio
 - reversing the order.
 - writing the monohybrid ratio instead of dihybrid ratio
- (c) Only a few candidates answered this sub part correctly. Many candidates mentioned the laws of segregation or dominance instead of independent assortment. The most common mistake found was trying to write the law in their own words, which simply could not capture the essence of the law.

Suggestions for teachers

- Reinforce definitions using comparison charts of excretion, egestion, and secretion.
- Use Venn diagrams comparing structure and function while teaching human physiology.
- Provide diagrams of various organ systems with incorrectly organised parts and ask students to identify and correct the errors.
- Conduct spelling drills and concept maps linking body systems.
- Encourage students to use appropriate key terms in definitions.
- Share visuals/posters/ news clippings or short films from Swachh Bharat with students.
- Guide students to list realistic objectives of Swachh Bharat Abhiyan in their own words based on their understanding.
- Use visual evolution timelines with feature comparisons. Create quizzes focusing on key physical traits of human ancestors.
- Emphasise genotype-phenotype links through Punnett square activities.
- Reinforce ratio patterns by providing the students with multiple solved examples.
- Clearly explain the offspring of a monohybrid and dihybrid cross.
- Discuss examples other than those given in the textbook and explain the offspring produced for better clarity in monohybrid and dihybrid crosses.
- Guide students to learn the laws put forth by Mendel only after understanding them well.
- For better retention, instruct the students to practice writing the laws regularly.

GENERAL COMMENTS

Topics found difficult by candidates

- Location of Ciliary Body and Suspensory Ligament.
- Sequence based questions.
- Functions of different types of white blood cells.
- Parts of the ear responsible for static equilibrium and dynamic equilibrium.
- Evolution of man- characteristic features of Cro magnon man, industrial melanism.
- Definition of terms - Micturition, Excretion, Allele.
- Mendel's Laws of Inheritance - Law of Independent Assortment. Many candidates could not articulate the law clearly.
- Defects of the eyes and their causes.
- Reason for throat infection leading to ear infection.
- Part of Autonomic Nervous System causing an increase in the secretion of Adrenaline.
- Diagram-based questions like plasmolyzed cell and the photosynthesis setup.
- Stages of mitosis.
- Difference between centrosome and centromere.
- Assertion-Reason type of questions.
- Candidates could not apply their learning to a new context in the MCQ on mice with cut tails.

Concepts in which candidates got confused

- Mortality/ Natality.
- Function of Seminal vesicle / Prostate Gland.
- Turgor and wall pressure acting in opposite directions.
- Characteristic features of different stages of Mitosis.
- Photosensitive Cells of the eyes and their pigments.
- Hormones and the disorders caused due to over and under secretion of the hormone.
- Function of Coronary artery / Pulmonary Artery, Inguinal Canal, Sperm Duct, Pinna / Auditory Canal, Urethra / Ureter.
- Location of plasma membrane, cytoplasm and hypertonic solution in case of a plasmolyzed cell.
- Cause of Cataract – opacity of Lens or Cornea.
- Functions of amniotic fluid and placenta.
- Some candidates believed caffeine stimulates the heart or nervous system directly rather than the brain.
- Functions of urethra and uterus.
- Dynamic and static equilibrium.

Suggestions for candidates

- Read and adhere to the scope of the prescribed syllabus.
- Study all chapters as per syllabus; do not skip any topic.
- Compare closely related topics to understand distinctions and similarities.
- Revise topics repeatedly to strengthen concept retention.
- Practice writing the spellings of biological terms to eliminate errors.
- Make a list of all biological abbreviations related to the syllabus.
- Understand diagrams thoroughly before attempting diagram-based questions.
- Practice neat, well-labelled diagrams regularly.
- Perform simple experiments to develop conceptual clarity and observation skills.
- Record data and analyse results accurately in practical notebooks to enhance understanding.
- Solve more multiple-choice questions, especially Assertion and Reason types.
- Understand and learn Mendel's laws of heredity.
- Use the 15-minute reading time effectively to plan your answers.
- Follow all instructions for each question carefully and answer accordingly.
- Avoid rushing to conclude an answer; write thoughtfully and precisely.
- Always write the correct question number before each answer.
- Ensure your handwriting is clear and legible.
- Write answers in short, simple sentences using correct scientific terms.
- Use appropriate prepositions when describing the location and function of structures/organs.
- Allocate the last 10 minutes to revise and eliminate any errors or omissions.
- Answer all questions as per the rubric specified in the paper.
- Write answers independently and confidently.